

Nguyen Thanh Duy Tan

AI Engineer Intern

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Career Objective

Short-term: Apply AI/ML knowledge to real projects and improve practical engineering skills.

Long-term: Become an AI Engineer capable of building reliable ML and LLM-based systems.

Education

Ho Chi Minh City University of Natural Resources and Environment Oct 2023 – Present
Major: Information Technology GPA: 3.0/4.0

Achievements

Academic Encouragement Scholarship – Good Classification, Academic Year 2024–2025.

Skills

- **Machine Learning:** Scikit-learn, Pandas, NumPy, Imbalanced Data Handling, EDA.
- **Generative AI & NLP:** LLMs, LangGraph, LangChain, Multi-Agent Systems, RAG.
- **MLOps & Software Engineering:** FastAPI, Docker, CI/CD (GitHub Actions), RESTful APIs.
- **Database & Storage:** MongoDB (Motor Async).
- **Tools:** Git, GitHub, Jupyter Notebook.
- **Soft Skills:** Problem-solving, system design thinking, self-researching technical docs.

Projects

VietLex – Vietnamese Legal RAG System 07/2026 – Present

- Developing a Vietnamese legal RAG system for a pinned corpus of 518,255 documents, with resumable ingestion, integrity checks, compressed full-text storage in SQLite/Zstandard, and document-level vectors in Pinecone Serverless.
- Implemented concurrent Pinecone dense/sparse retrieval and SQLite FTS5 lexical search, followed by legal-structure-aware chunking and remote ColBERT reranking with a Pinecone BGE reranker fallback.
- Added revision-aware semantic caching, NeMo input/output guardrails with PII redaction, provider timeout and fallback handling, Logfire tracing, and an evaluation harness for factoid, multi-hop, and unanswerable queries.

Tech Stack: Python, FastAPI, Pinecone, Qdrant Cloud, SQLite FTS5, E5-small, ColBERT, NeMo Guardrails, Logfire, Docker.

GitHub: github.com/TanNguyen234/VietLex-Tech-Spec

CV AI Evaluation System – Multi-Agent LLM Application 03/2026 – 05/2026

- Built a LangGraph workflow for PDF extraction, candidate profiling, dynamic rubric generation, Tavily-based market enrichment, and parallel execution of six specialized evaluator nodes.
- Implemented deterministic cross-agent validation, meta-evaluation, optional job-description matching, and retry policies for transient LLM failures across configurable Gemini and Qwen providers.
- Developed an async FastAPI service with SSE progress streaming, MongoDB/Motor persistence, LLM-based CV validation, per-IP daily limits, unit/integration tests, and Docker deployment on Hugging Face Spaces.

Tech Stack: Python, FastAPI, LangGraph, LangChain, Gemini, Qwen, Tavily, MongoDB/Motor, PyMuPDF4LLM, Docker.

GitHub: github.com/TanNguyen234/CV-Review-System | **Demo:** foxy0505-ai-cv-reviewer.hf.space/app

Fraud Detection API & CI/CD Pipeline 05/2026 – 06/2026

- Trained an XGBoost classifier on the ULB credit-card dataset (approximately 0.17% fraud) using duplicate removal, time transformation, RobustScaler, stratified splitting, and class weighting with `scale_pos_weight`.
- Evaluated the model with PR-AUC and class-wise metrics, then served the preprocessing and model artifacts through versioned FastAPI/Pydantic endpoints for single and batch inference of up to 100 transactions.
- Added unit/integration tests and a GitHub Actions workflow that runs Ruff and Pytest before building and pushing the Docker image to Docker Hub.

Tech Stack: Python, XGBoost, Pandas, Scikit-learn, FastAPI, Pydantic, Pytest, Docker, GitHub Actions.

GitHub: github.com/TanNguyen234/End-to-End-Fraud-Detection-Pipeline